

Preliminary Exam in Analysis, May 2022

Problem 1. Let $[0, 1] \subset \mathbb{R}$ be the unit closed interval. For a continuous function $f : [0, 1] \mapsto \mathbb{R}$ and $0 < \alpha < 1$ we define

$$[f]_\alpha := \sup_{x, y \in [0, 1], x \neq y} \frac{|f(x) - f(y)|}{|x - y|^\alpha},$$

and

$$\|f\|_\alpha := \sup_{x \in [0, 1]} |f(x)| + [f]_\alpha.$$

We now define

$$X_\alpha := \{f \in C^0([0, 1]); \|f\|_\alpha < +\infty\}; \quad \forall f, g \in X_\alpha \quad d_\alpha(f, g) := \|f - g\|_\alpha.$$

- (a) Prove that the metric space (X_α, d_α) is complete. You do not need to prove that d_α is a metric.
- (b) Let f_k be a bounded sequence in (X_α, d_α) . Prove that there exists a subsequence of f_k which is uniformly converging. Prove moreover that the limit belongs to X_α .
- (c) Let $f_0(x) = \sqrt{x}$. Prove that $f_0 \in X_\alpha$ if and only if $0 < \alpha \leq 1/2$.

Problem 2. Let (X, d) be a metric space and let $K \subset X$ be a compact set. Let for all $x \in X$:

$$d(x, K) := \inf_{z \in K} d(x, z)$$

Prove that

$$\forall x, y \in X \quad |d(x, K) - d(y, K)| \leq d(x, y).$$

Problem 3. Let S be the unit sphere in $\mathbb{R}^3$ and let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be a C^3 function which vanishes on S . Assume that for a constant $c > 0$

$$f(x) \geq c \operatorname{dist}^2(x, S)$$

where

$$\operatorname{dist}(x, S) := \inf_{y \in S} |x - y|.$$

Prove that for all $x_0 \in S$, $v \in \mathbb{R}^3$, we have

$$v \cdot D^2 f(x_0) v \geq 2c |v \cdot x_0|^2,$$

where $D^2 f = [\frac{\partial^2 f}{\partial x_i \partial x_j}]$ is the Hessian matrix of the 2nd derivatives.

Problem 4. (1) Let $\vec{F}$ be a smooth vector field in $\mathbb{R}^n$. Let $B^n(x_0, r)$ denote the ball centered at $x_0 \in \mathbb{R}^n$ with radius $r > 0$, let $S^{n-1}(x_0, r)$ denote the sphere centered at x_0 of radius r , an let $\vec{n}$ be the outer unit normal in $S^{n-1}(x_0, r)$. Let $|B^n(x_0, r)|$ denote the n -dimensional volume of the ball $B^n(x_0, r)$. Prove that we have

$$(\operatorname{div} \vec{F})(x_0) = \lim_{r \rightarrow 0} \frac{1}{|B^n(x_0, r)|} \int_{S^{n-1}(x_0, r)} \langle \vec{F}(y), \vec{n}(y) \rangle d\sigma(y).$$

- (2) Given fixed unit vector $\nu \in \mathbb{R}^3$, let $D(x_0, r)$ be the 2-dimensional disk centered at x_0 with radius r and perpendicular to ν . Let $\vec{t}$ be the unit tangent vector to $\partial D(x_0, r)$. Prove that we have

$$\langle (\operatorname{curl} \vec{F})(x_0), \nu \rangle = \lim_{r \rightarrow 0} \frac{1}{\pi r^2} \int_{\partial D(x_0, r)} \langle \vec{F}(y), \vec{t}(y) \rangle ds(y).$$

Problem 5. Let $f: [a, b] \mapsto \mathbb{R}$ be a Riemann integrable function. Define the coefficients

$$a_n(f) = \int_a^b f(x) \sin(nx) dx.$$

Show that

$$\lim_{n \rightarrow \infty} a_n(f) = 0.$$

HINT: First prove it for the characteristic function of an interval $[\alpha, \beta] \subset [a, b]$. Then prove it for a (finite) linear combination of characteristic functions of intervals (these are called simple functions). Since f is Riemann integrable, given $\epsilon > 0$ use a lower sum to show that f can be approximated by a simple function

$$g = \sum_{\text{finite}} c_i \chi_{I_i} \leq f$$

for certain numbers c_i and intervals I_i in the following sense

$$0 \leq \int_a^b (f - g) dx < \epsilon.$$

Deduce the statements for f from the statements for g .

Problem 6. Let $f: \mathbb{R}^n \mapsto \mathbb{R}$ be a C^1 function such that $f(0) = 0$. Let $G: \mathbb{R}^n \mapsto \mathbb{R}^n$ be a C^1 mapping such that $G(0) = 0$ and $DG(0)$ is invertible. Prove that there exists an open neighborhood U of the origin in $\mathbb{R}^n$ and a continuous mapping $H: U \mapsto \mathbb{R}^n$ such that

$$f(x) = \langle G(x), H(x) \rangle = \sum_{i=1}^n G^i(x) H^i(x)$$

for every $x \in U$.

HINT: Use the inverse function theorem to reduce the problem to showing that any function h which is C^1 and satisfies $h(0) = 0$ can be written in the form

$$h(y) = \langle y, K(y) \rangle,$$

for a continuous mapping K defined in a neighborhood of zero.